

Talha Ahmed

Pipeline & Automation Engineer · Technical Artist

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SUMMARY

Pipeline and tools engineer and technical artist, **16+ years** building production software, tools, and automation for animation, VFX, and games. Deep in open-standard asset pipelines (**OpenUSD, MaterialX / OpenPBR**) and large-scale, cloud-backed automation, increasingly built with **AI / LLM tooling**. Ships clean, tested **Python** and **C++** across DCCs (**Houdini, Maya, Nuke, Unreal**); remote and async across timezones.

TECHNICAL SKILLS

Languages	Python (primary), C/C++, C#, JavaScript, Lua, Bash, PowerShell, MEL/VEX
DCC & CG	Houdini (Solaris, HDA/OTL, hython, PDG/TOP), Maya, Nuke, Unreal (Interchange), Blender, Arnold, RenderMan, Deadline; OpenUSD, MaterialX, OpenPBR, OpenImageIO
Cloud & Tooling	AWS (S3, EC2, Step Functions), FastAPI, CI/CD, pytest, ruff, uv, Rez, Qt/PySide2, Perforce, Git, ShotGrid/fftrack
ML / AI	LLMs & agents (Claude Code), Stable Diffusion, TensorFlow, PyTorch, Hugging Face

EXPERIENCE

Epic Games — Quixel — *Pipeline & Automation Engineer* Remote · Apr 2025 – Jul 2026

- Core developer and architect of a **dual-backend** 3D-asset conversion pipeline on **OpenUSD + MaterialX / OpenPBR** — a **Houdini** engine (Solaris, custom HDAs, hython) driven from a clean Python API, plus a **pure-Python** backend with no DCC dependency, kept in parity by an automated **QA gate**.
- Designed a **provider-based asset model** (schema validation + metadata auto-repair) and **corpus-scale** batch tooling for tens of thousands of legacy assets — corruption detection, auditable rejections, idempotent relaunch, 404-aware gather (**78.7%** → **98.9%** coverage).
- In-house **FBX→USD reader** (Autodesk FBX SDK, two-pass local-space reconstruction) preserving per-node pivots; shared color-space + OIIO displacement logic. Ran a defect→rule review loop with the receiving pipeline team; remote/async across US/UK/EU/Asia; built with **Claude Code / LLM agents**.

Blink Industries — *Pipeline Developer* London · Remote · Aug – Oct 2024

- Hybrid 2D/3D Maya / Toon Boom Harmony / Adobe pipeline: TD tools, Premiere edit generation, After Effects layout, 3D-object integration into Harmony.

Untold Studios — *Pipeline Developer* London · Remote · Oct 2023 – May 2024

- VFX-heavy Maya / Houdini / Nuke pipeline; improved artist UX in data ingestion and review publishing; requirements analysis with animation and match-move craft heads.

One Animation / Moonbug — *Sr Pipeline TD* Singapore · Remote · Sep 2022 – Sep 2023

- Lean Maya / Houdini pipeline: Sphinx auto-docs, costume/animation libraries, asset ingestion & archiving, config management with Rez.

Passion Pictures — *Pipeline Developer* London · Remote · Mar – Sep 2022

- Software-agnostic VFX/animation pipeline: Arnold rendering, Arnold+Nuke deep compositing, ShotGrid tool integration via Python API.

Nebula Pipeline — *Studio Developer / Consultant* Karachi · 2019 – 2021

- Built the **Nebula Pipeline Toolbox** — a documented Maya/Redshift production pipeline (planning → assets → layout/anim → lighting/render → post). [docs](#).

Earlier — *ICE Animations (Pipeline Supervisor / CG Researcher) · Post Amazers (Technical Artist)* Karachi · 2004 – 2019

- Led an agile team building a Maya/Nuke pipeline for feature/TV/TVC production; RenderMan/RSL and C/C++ tools, plugins, effects, and stereoscopy. Credits incl. *Mansour* (2012), *Son of the Mask* (2005).

EDUCATION & CERTIFICATIONS

- B.S. Computer Systems Engineering** — GIK Institute (1999–2003) · **EM CIMET** color-imaging coursework — U. Granada & Gjøvik UC (2008–2010) · **P.G.D. Business Admin** — IBA Karachi (2005–2007)
- Deep Learning Specialization** — DeepLearning.ai / Coursera (2019) · **AI & GPT Applications** — Encode.club (2024)